TWO OUT OF THREE FOR THE BLOW-UP SQUARE CONDITION ON A DISTINGUISHED TRIANGLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the closing 'Some details omitted; use Derived Categories, Lemma derived-lemma-third-isomorphism-triangle' names the general result that supplies the routine compatibility, and the sentence is preserved verbatim in the counted span.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The canonical map of a blow-up square. Same chapter, Lemma lemma-c-square. Supplies the maps whose being isomorphisms is the condition considered.
- Representation of the third term of a distinguished triangle by a cone. Derived Categories, Lemma derived-lemma-triangle-K-injective. Represents the third complex by the cone on a map of K-injective complexes.
- The third isomorphism in a map of distinguished triangles. Derived Categories, Lemma derived-lemma-third-isomorphism-triangle. Supplies the argument that the third leg is a quasi-isomorphism.

Source reference. Original lemma 090X: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6061–6061. Navigation pointer invoked at proof line 6061; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 014A: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6066–6066. Navigation pointer invoked at proof

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line 6066; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Consider a commutative diagram

$$\begin{array}{ccc} E & \longrightarrow & Y \\ \downarrow & & \downarrow \\ Z & \longrightarrow & X \end{array}$$

in the category $\mathcal{C}$. In this situation, given an object K of $D(\mathcal{O})$ we get what looks like the beginning of a distinguished triangle

$$R\Gamma(X, K) \to R\Gamma(Z, K) \oplus R\Gamma(Y, K) \to R\Gamma(E, K)$$

In the following lemma we make this more precise.

Lemma 0.1. *In the situation above, let $K_1 \to K_2 \to K_3 \to K_1[1]$ be a distinguished triangle in $D(\mathcal{O})$. If $c^{K_i}_{X,Z,Y,E}$ is a quasi-isomorphism for two i out of $\{1, 2, 3\}$, then it is a quasi-isomorphism for the third i.*

Proof. By rotating the triangle we may assume $c^{K_1}_{X,Z,Y,E}$ and $c^{K_2}_{X,Z,Y,E}$ are quasi-isomorphisms. Choose a map $f : \mathcal{I}_1^\bullet \to \mathcal{I}_2^\bullet$ of K-injective complexes of $\mathcal{O}$-modules representing $K_1 \to K_2$. Then K_3 is represented by the K-injective complex $C(f)^\bullet$, see Derived Categories, Lemma 090X. Then the morphism $c^{K_3}_{X,Z,Y,E}$ is an isomorphism as it is the third leg in a map of distinguished triangles in $K(Ab)$ whose other two legs are quasi-isomorphisms. Some details omitted; use Derived Categories, Lemma 014A. □